you can subscribe using the multi-level wildcard # to get data on all topics), you will see a deluge of messages of all types that people all over the world are publishing to this free broker.

Listing 3.2 A simple MQTT publish and subscribe example using a public MQTT broker.

```
import paho.mqtt.client as mqtt
import time

BROKER = "test.mosquitto.org"  #1
TOPIC = "home/environment/bedroom1/temp_sensor_1"

def on_message(client, userdata, message): #2
  print(f"Message received: {message.payload} on topic {message.topi
c}")

client = mqtt.Client(mqtt.CallbackAPIVersion.VERSION2) #3
client.on_message = on_message
client.connect(BROKER)

client.subscribe(TOPIC) #4
client.loop_start() #5

for i in range(5): #6
  temp = 20 + i
  client.publish(TOPIC, json.dumps({"temperature": temp}))
  time.sleep(2)

client.loop_stop() #7
client.disconnect()
```

#1 Use a free public MQTT broker.
#2 Define the message handler to run when a message is received.
#3 Create a new MQTT cleint, define the message handler, and connect to the broker.
#4 Change this to be "#" to see all data on this broker.
#5 Start the client listening.
#6 Publish 5 messages to the topic.
#7 Cleanup.

3.4 Real world data collection strategies

Not every aspect of the physical world can be captured by electronic sensors. Despite the proliferation of IoT devices and ever-decreasing sensor costs,